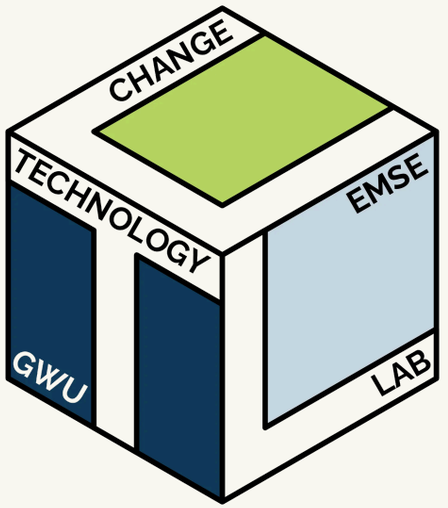
Quantifying the Peak-Shaving Potential and Cost Efficiency of BEV Supplier-Managed Charging

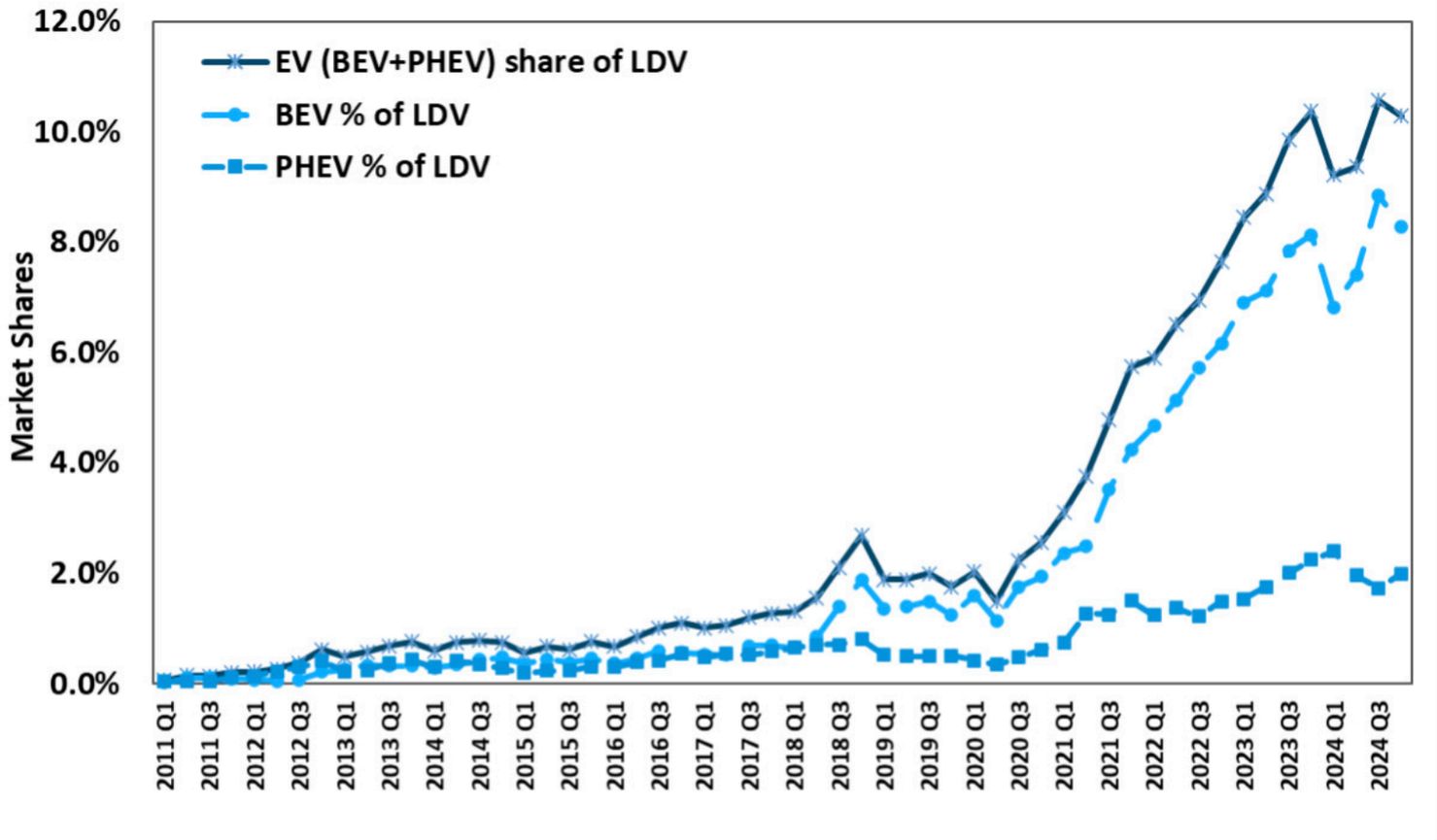
Integrating consumer enrollment with grid simulation in CAISO and NYISO



Pingfan Hu
George Washington University

with Brian Tarroja, Matthew Dean, Kate Forrest,
Eric Hittinger, Alan Jenn, and John Paul Helveston

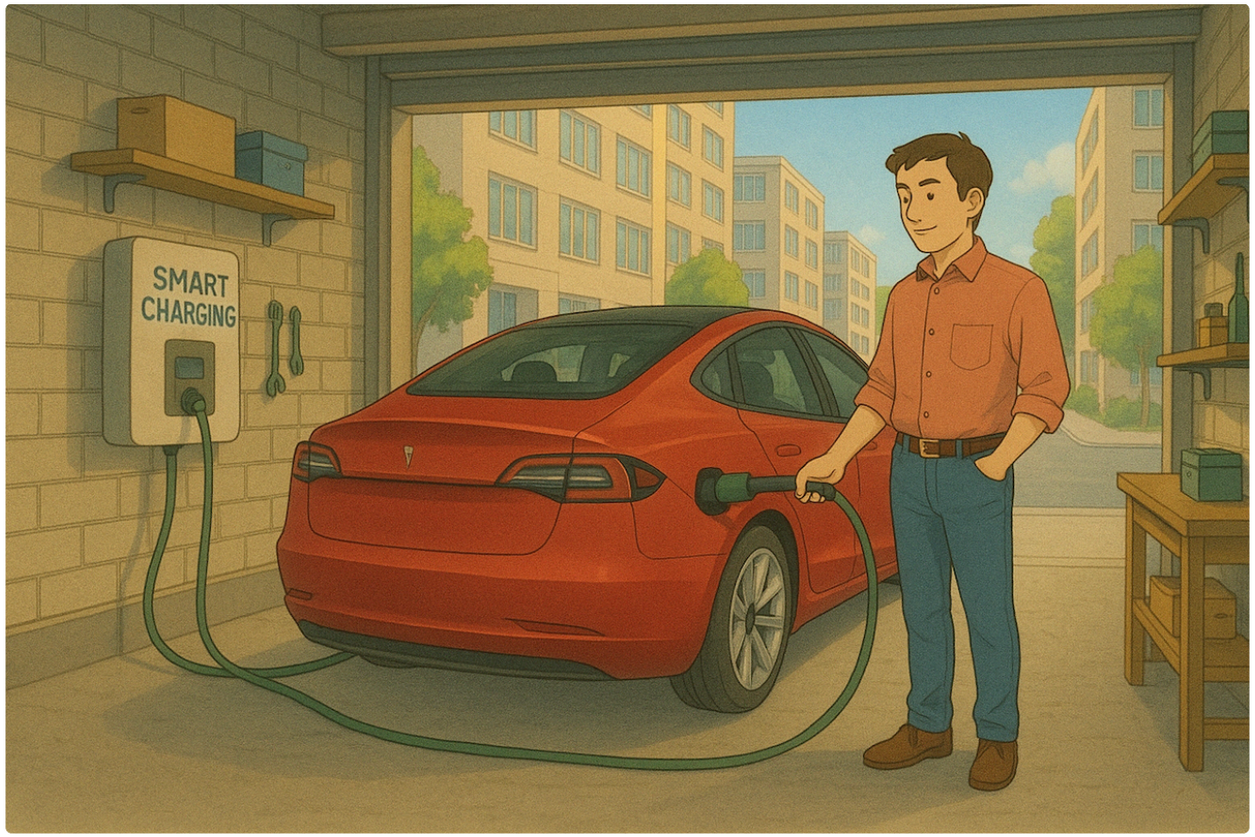
EV sales in US reaching ~10% of sales



Source: Argonne National Lab, www.anl.gov/ev-facts/model-sales

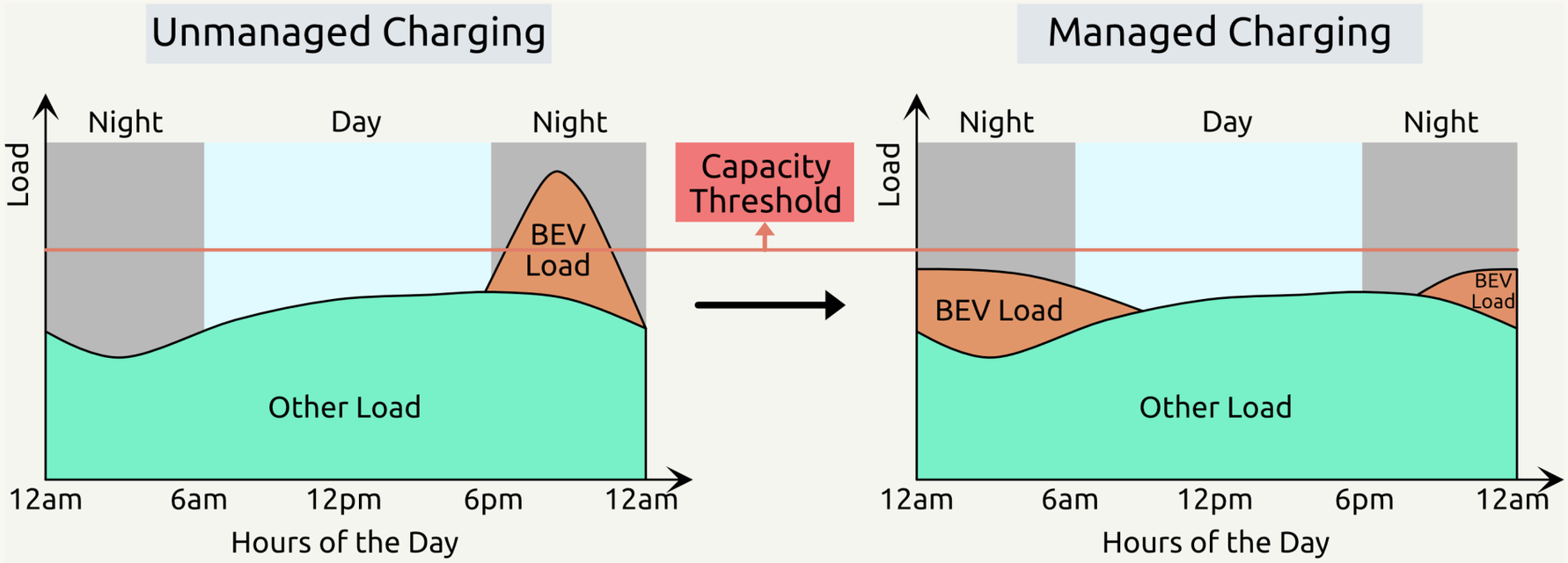
Introduction

- **Unmanaged** BEV charging is becoming a problem to the grid.
- **Managed** charging is cheaper and smooths out the grid load.
- **Smart** charging: Supplier-Managed Charging (SMC) and Vehicle-to-Grid (V2G).



SMC - Supplier Managed Charging

- SMC smooths out the BEV charging spike (called “Peak-shaving”).
- Electricity demand is controlled below the capacity threshold.
- It saves money and reduces pollution.



Managed charging avoids overload caused by BEV charging.

Smart charging effectiveness depends on enrollment and peak-shaving.

PRIOR STUDY

Enrollment

Will BEV owners opt in,
and at what incentive?

THIS STUDY

Peak-shaving

How much peak can SMC shave,
and at what cost?

Prior study: willingness to enroll in SMC

Conjoint survey of BEV owners (Hu et al., 2025)

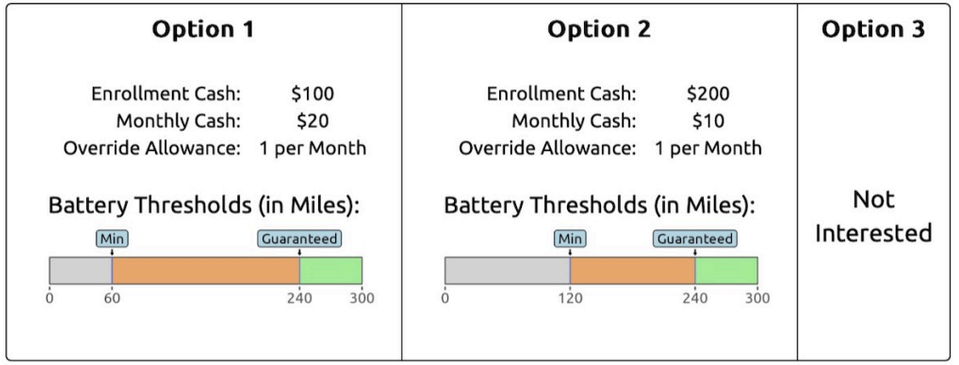
1,356
BEV owner responses

803 + 553
Meta Ads + Dynata, 2024

5 attributes
cash, overrides, battery thresholds

(1 of 6) If your utility offers you these 2 SMC programs, which one do you prefer?
(Your BEV has maximum range of **300** miles.)

Access the SMC Attributes

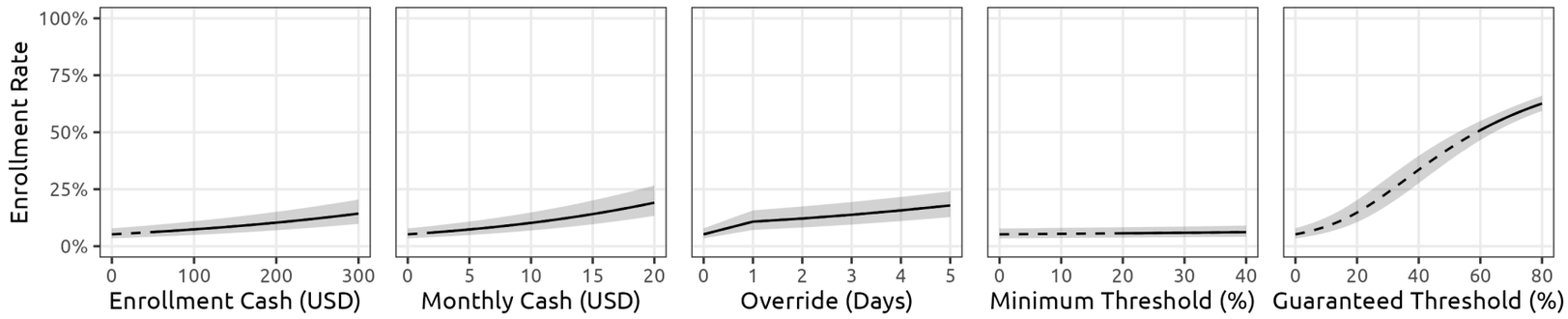


Prior study: SMC conjoint attributes

No.	Attribute	Range
1	Enrollment Cash	$50 to $300
2	Monthly Cash	$2 to $20
3	Monthly Override	0 to 5
4	Min Battery	20% to 40%
5	Guaranteed Battery	60% to 80%

The enrollment curve is a **model prediction** at a baseline profile, not a surveyed level: monthly cash varies, override = 0/month, thresholds = 0% / 40%.

Enrollment sensitivity of SMC features



1

Slope = sensitivity

Steeper curve, stronger response

2

Strongest levers

Monthly cash and guaranteed threshold

3

Diminishing returns

Each increment buys less enrollment

Prior grid studies assume enrollment is high and free

1. Most grid studies **assume** optimal or near-100% participation (Coignard et al., 2018; Tarroja et al., 2015), but real pilots enroll far fewer.
2. Enrollment is **not free**: utilities must pay incentives (Bailey et al., 2023; Wong et al., 2023), yet costs are rarely counted.
3. Few studies compare peak-shaving **across grid regions** with different load shapes.

We connect **consumer enrollment** with **grid simulation**.

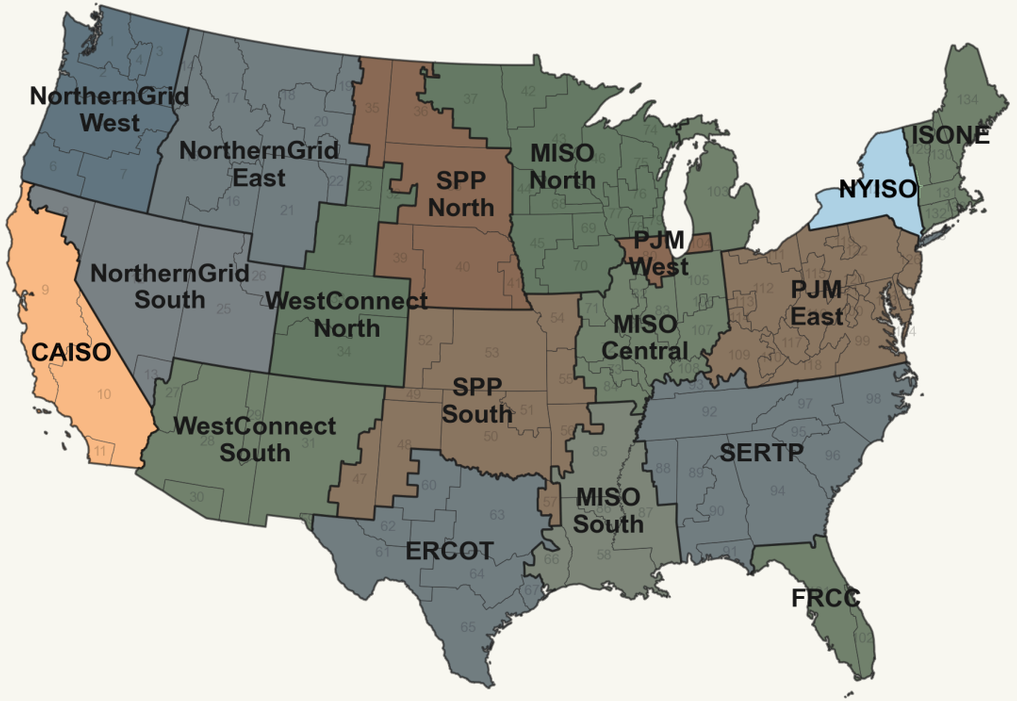
Research Questions

1. **Peak-shaving:** How much can SMC reduce peak demand in contrasting grid regions?
2. **Cost efficiency:** How much does peak shaving cost per unit, and is more enrollment always better?

Approach

1. **Scenario-based simulation** for CAISO and NYISO.
2. **Enrollment curve** from the prior study: incentives vs participation.

Two contrasting regions: CAISO and NYISO



	Single families	Energy source	Net load shape	Seasonal effect
CAISO	7.69 million	Solar	"Duck curve"	Minimal
NYISO	3.16 million	Hydro	Flat profile	Strong in winter

Both **lead in BEV adoption**, with **contrasting grid structures.**

Data Sources

Source	Provides	Role in Simulation
2022 NHTS	Real trip records nationwide	Charge time & energy needs
2025 GridStatus.io	Net load time series	Regional grid baselines
2025 U.S. Census	Single-family counts	Scale to total fleet energy
Prior study	Enrollment curve	Cost efficiency analysis



Vehicle & Charging Assumptions

BEV RANGE

200 miles

CHARGING

Level 2 (6.48 kW)

BATTERY SOC

20% – 80%

TRAVEL NEEDS

Guaranteed

OVERRIDES

2 – 3 per month

Simulation Overview

Scenario Inputs

Regions
CAISO
NYISO

Time Periods
Annual average
Extreme days
Seasonal

EV:House
1:1
1:2
1:4

SMC Enrollment
0%
10%
...
100%

Load Shifting Window
10AM-8PM -> 8PM-7AM
10AM-9PM -> 9PM-7AM
10AM-10PM -> 10PM-7AM
10AM-11PM -> 11PM-7AM
10AM-12AM -> 12AM-7AM

Peak-Shaving Model

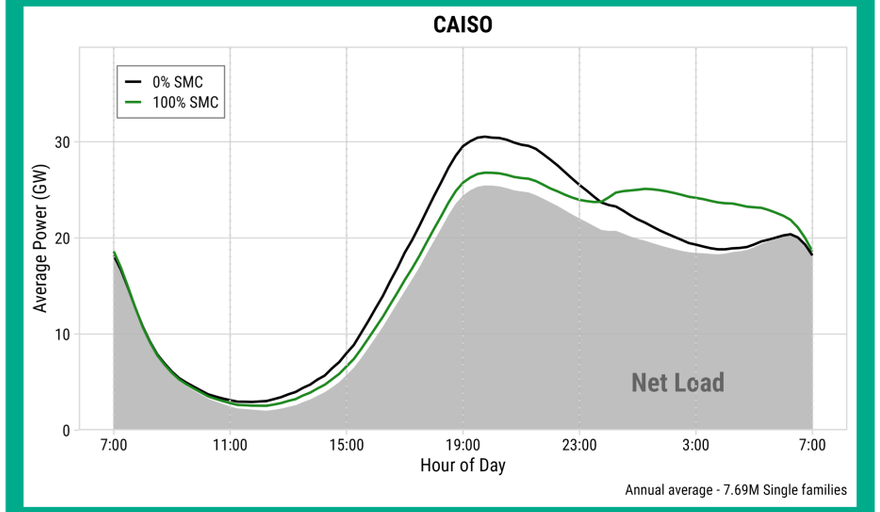
Data Source

Grid Net Load
2025 CAISO & NYISO time-series data

Household Count
2025 U.S. Census
CAISO: 7.69M
NYISO: 3.16M

Travel Patterns
2022 NHTS Results

One Example Run



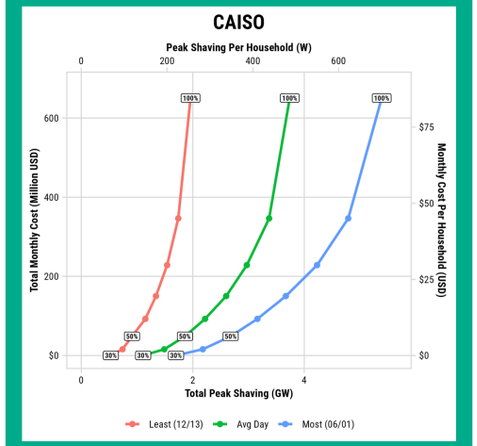
Run the peak-shaving model multiple times with varying scenario inputs.

Cost Efficiency Simulation

Data Source

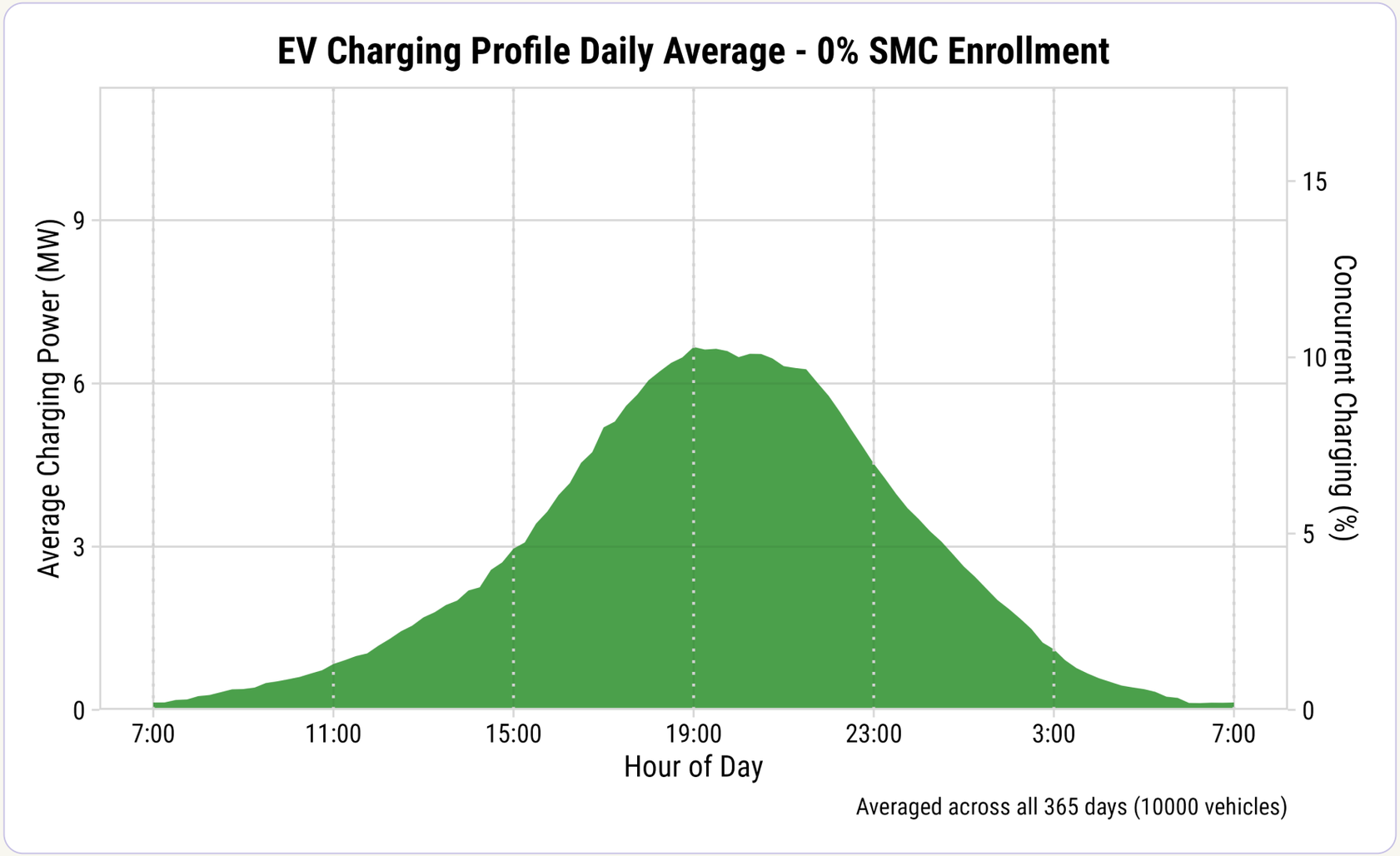
Enroll vs Cost Model
2024 Discrete Choice Experiment
(1,356 BEV owners)

Extreme Days Example



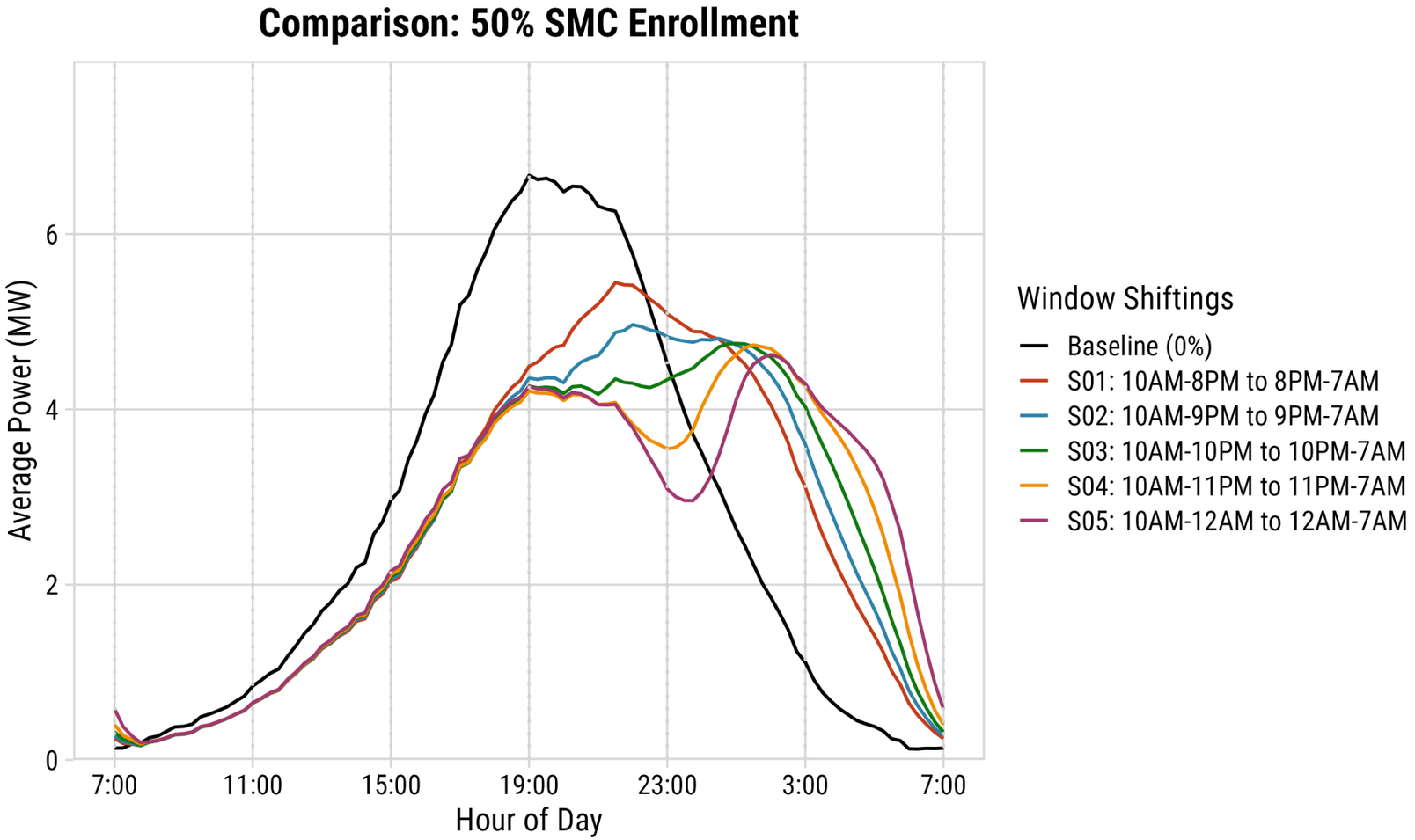
Contains annual average, extreme days, seasonal averages, and different EV-to-house ratios

Step 1: BEV Charging Load Profile



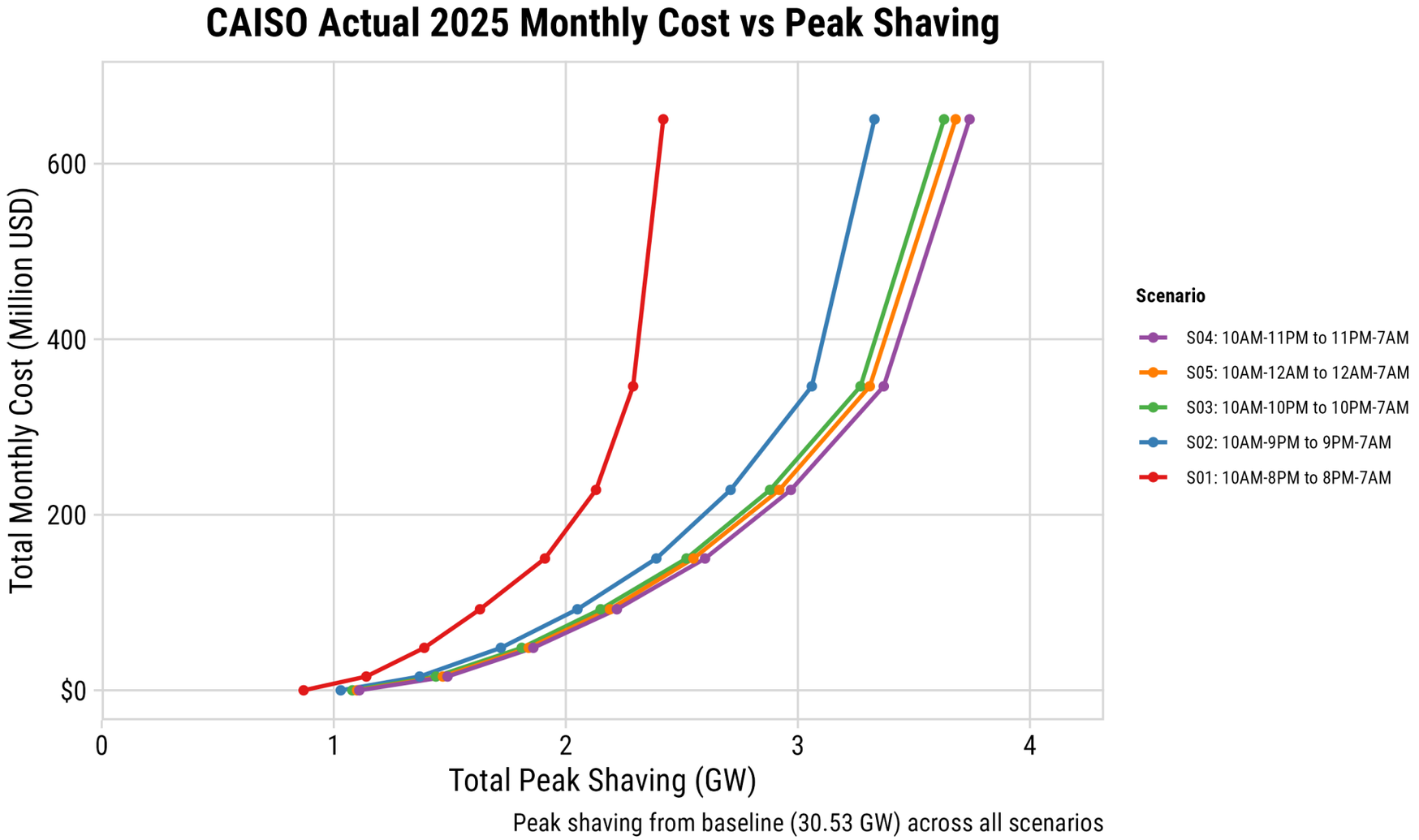
Simulation of 10,000 vehicles based on ~26,000 trip records from 2022 NHTS.

Step 2: BEV Load Peak Shaving

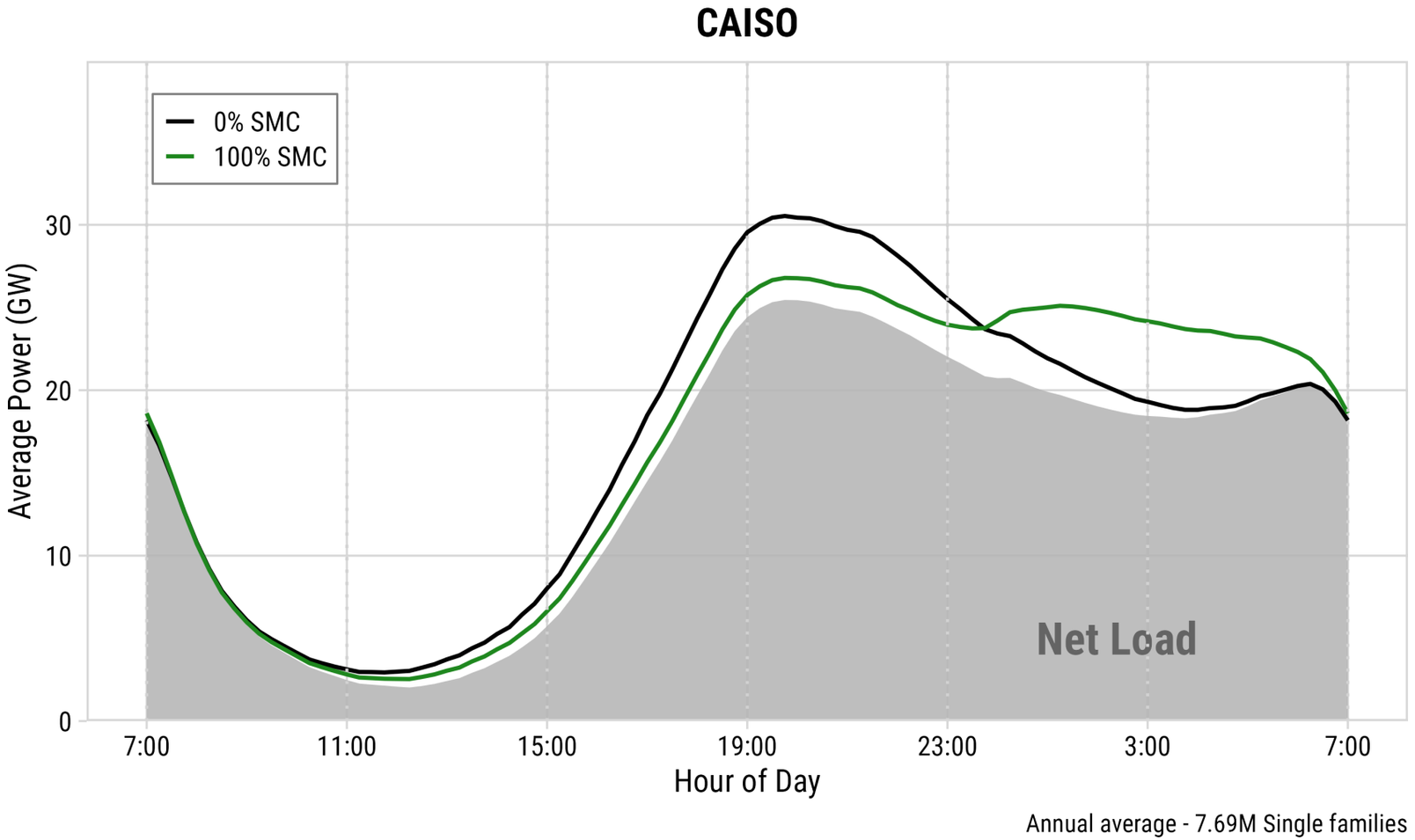


Five configurations with different peak/valley windows.

TOU window: 11 PM valley start performs best

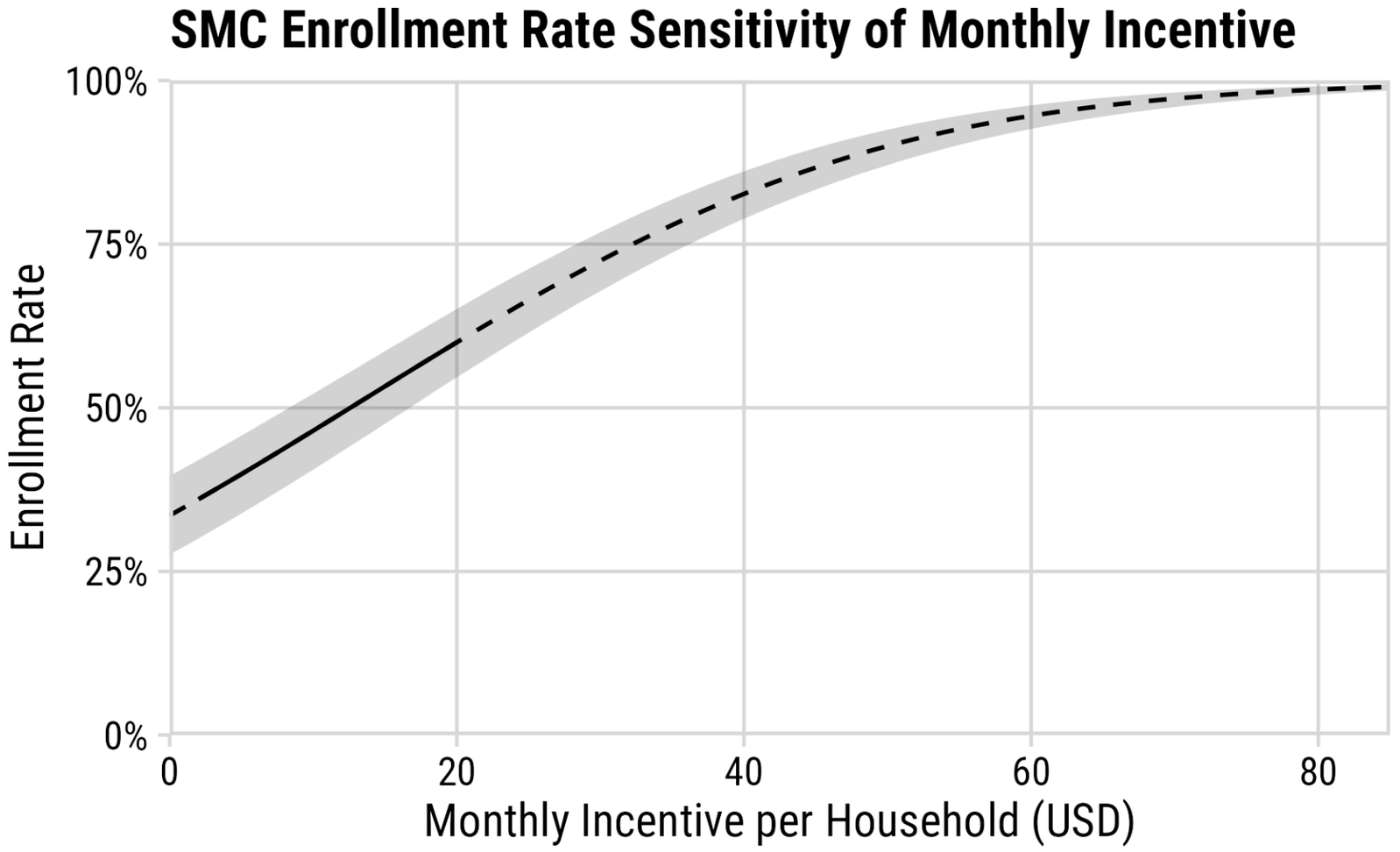


Step 3: Net Load + BEV & Peak Shaving



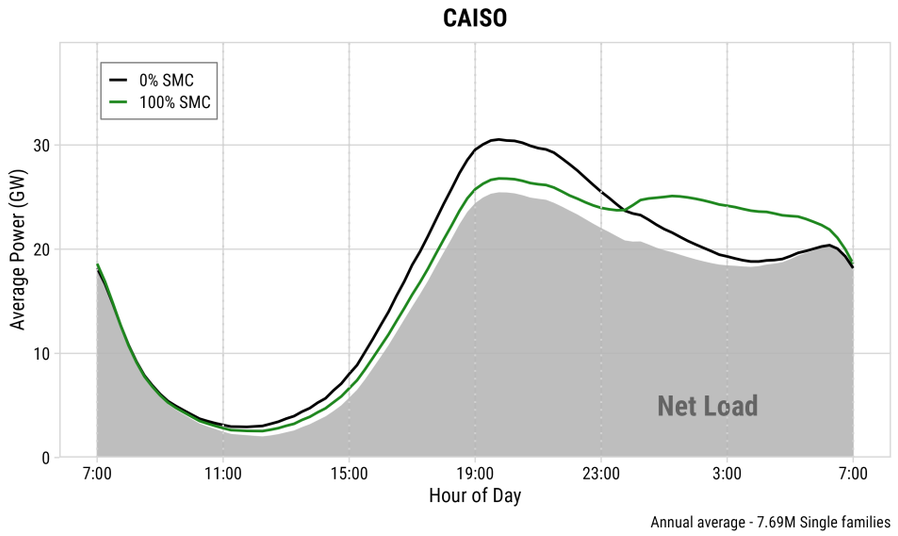
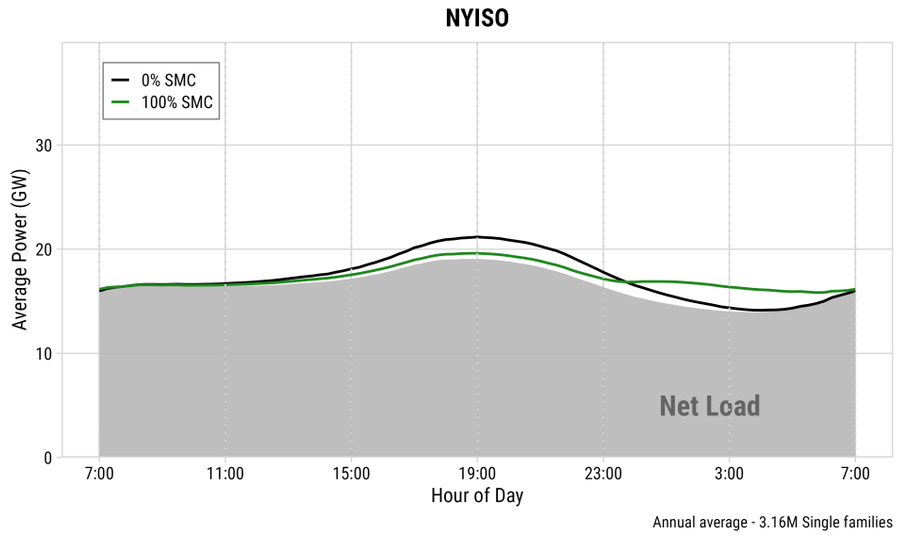
BEV load scaled to fleet size and stacked on net load.

Enrollment Curve from Prior Study



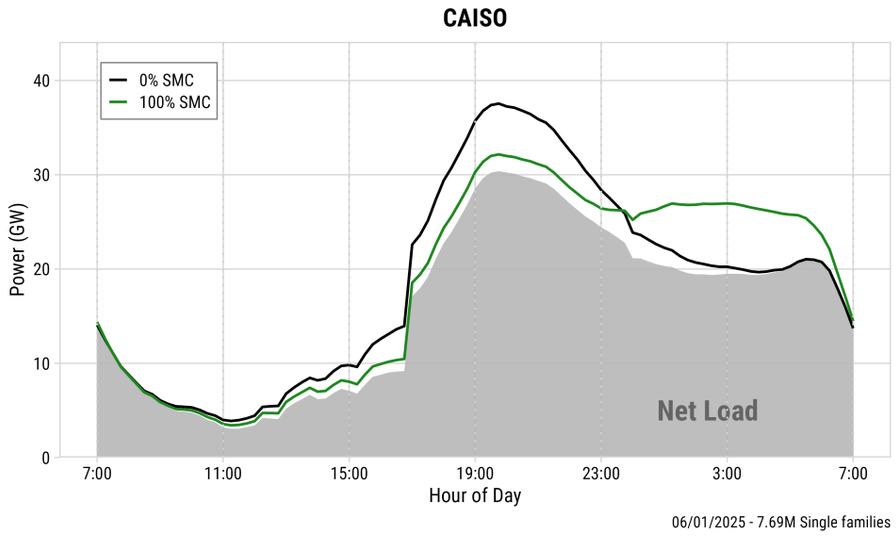
Mixed logit prediction based on 1,356 real BEV owners.

Peak Shaving: Annual Average

CAISO		NYISO	
			
Peak shave	3.7 GW (30.5 → 26.8)	**Peak shave**	1.6 GW (21.2 → 19.6)
Percentage	12.1%	**Percentage**	7.5%

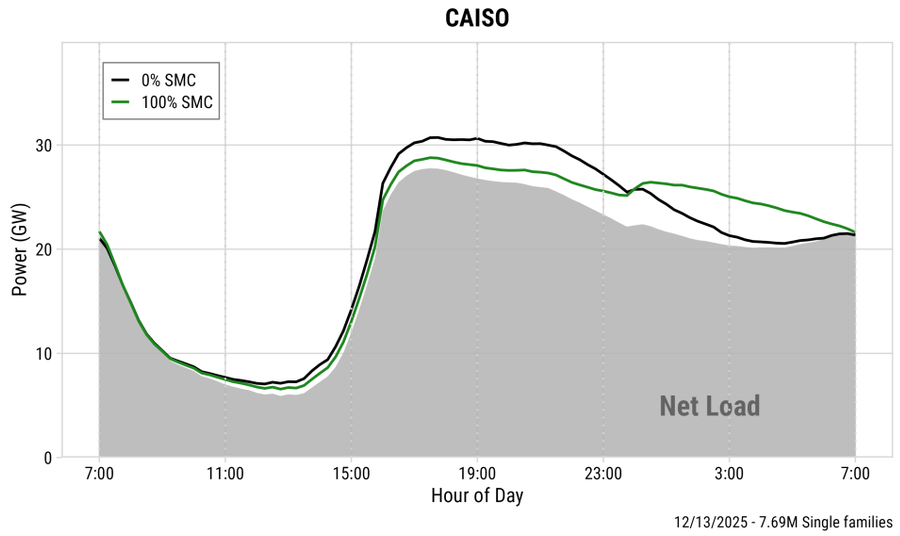
Peak Shaving: CAISO Extreme Days

Best Day (Jun 1)



Peak shave	5.4 GW (37.6 → 32.2)
Percentage	14.4%

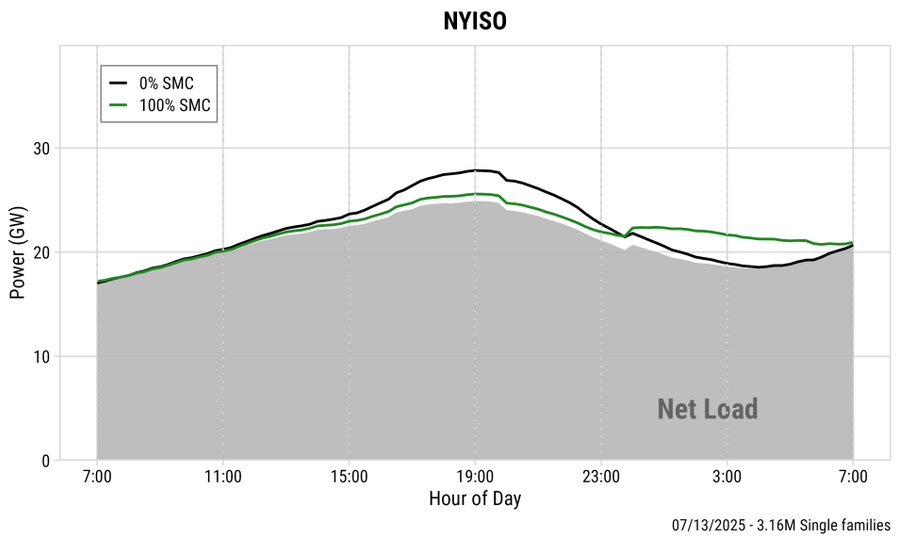
Worst Day (Dec 13)



Peak shave	1.9 GW (30.7 → 28.8)
Percentage	6.2%

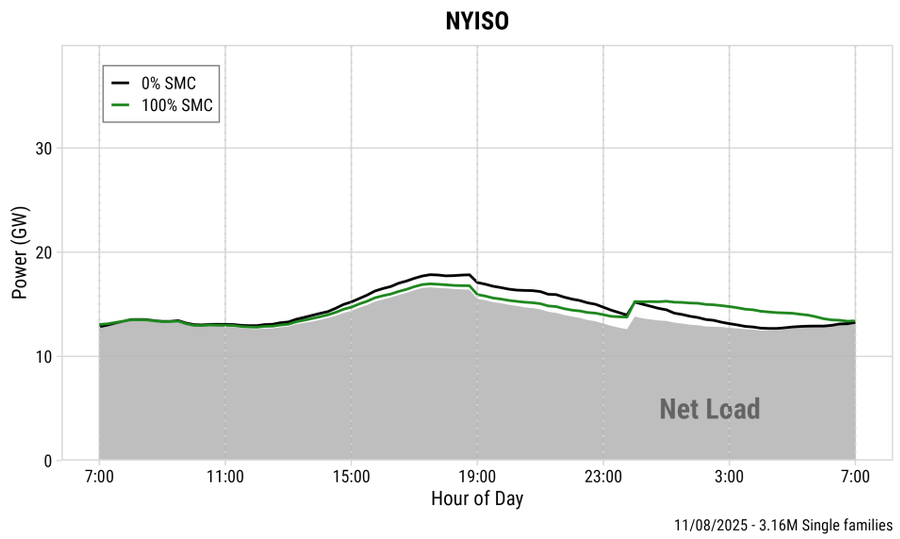
Peak Shaving: NYISO Extreme Days

Best Day (Jul 13)



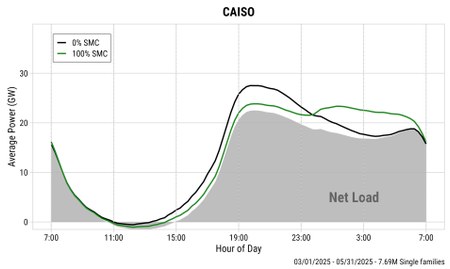
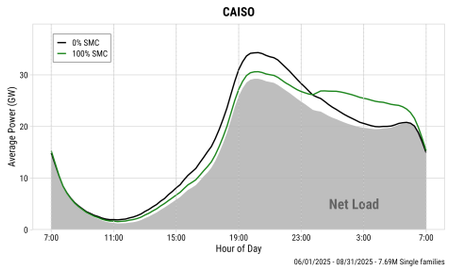
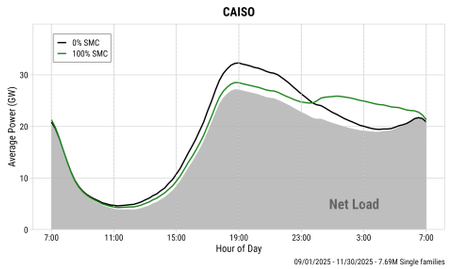
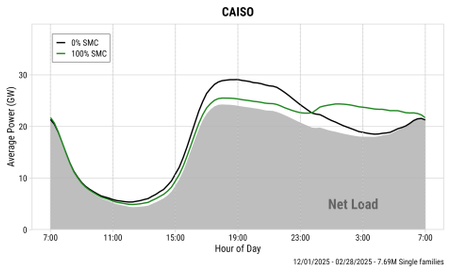
Peak shave	2.2 GW (27.8 → 25.6)
Percentage	7.9%

Worst Day (Nov 8)



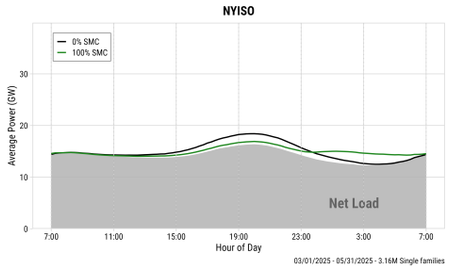
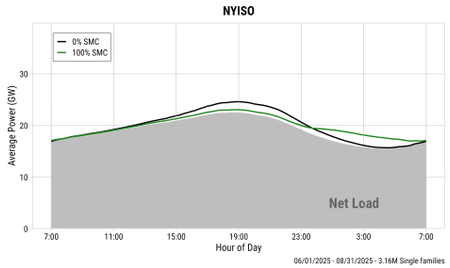
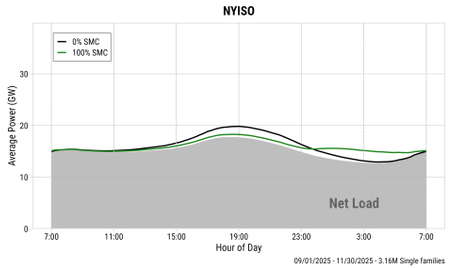
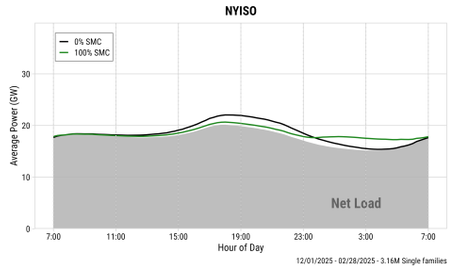
Peak shave	0.8 GW (17.8 → 17.0)
Percentage	4.5%

Seasonal Patterns: CAISO

Spring	Summer	Fall	Winter
			
3.7 GW (27.5 → 23.8)	3.7 GW (34.3 → 30.6)	3.8 GW (32.3 → 28.5)	3.6 GW (29.1 → 25.5)
13.5%	10.8%	11.8%	12.4%

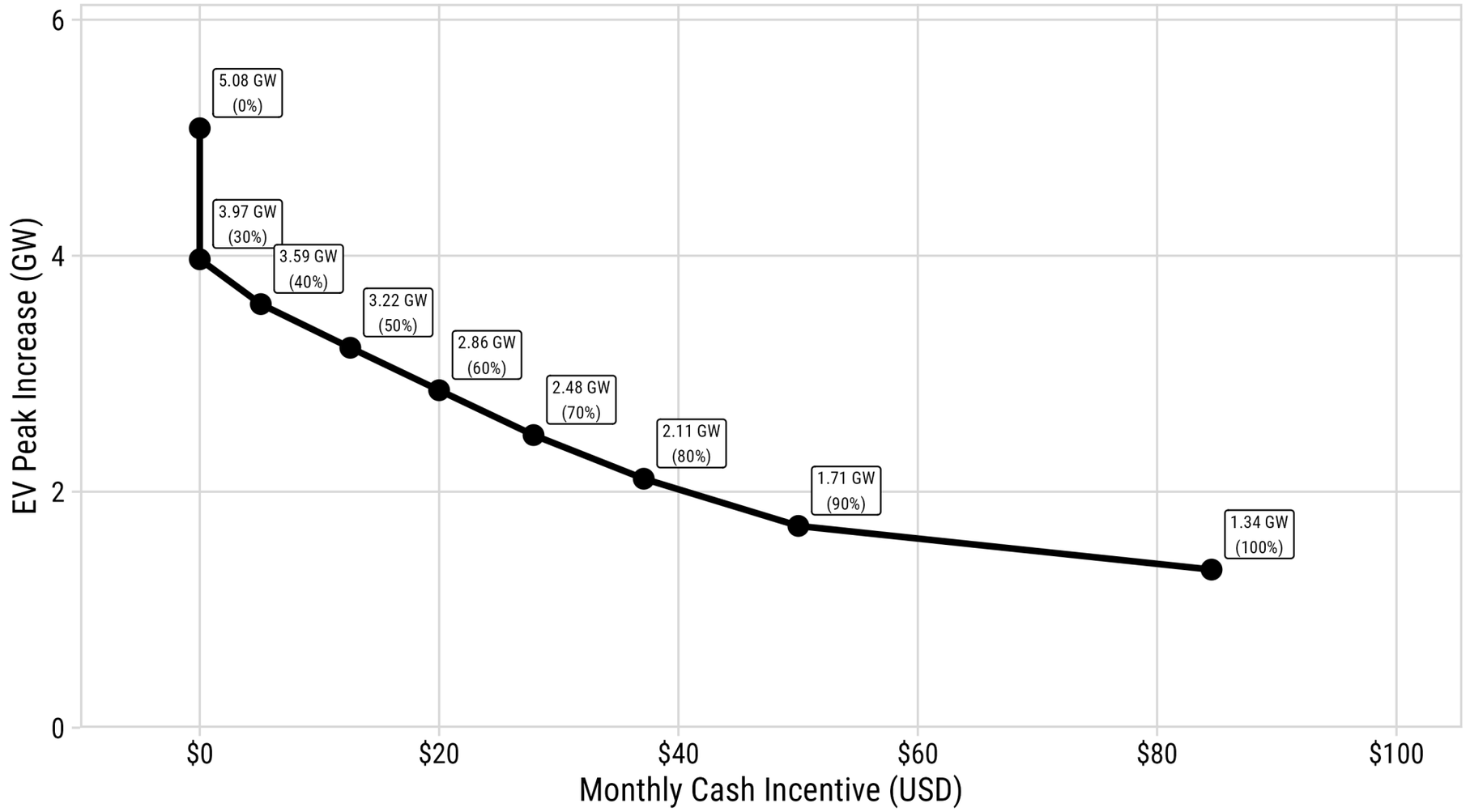
Peak shave for all seasons: **3.6–3.8 GW (10.8–13.5%)**

Seasonal Patterns: NYISO

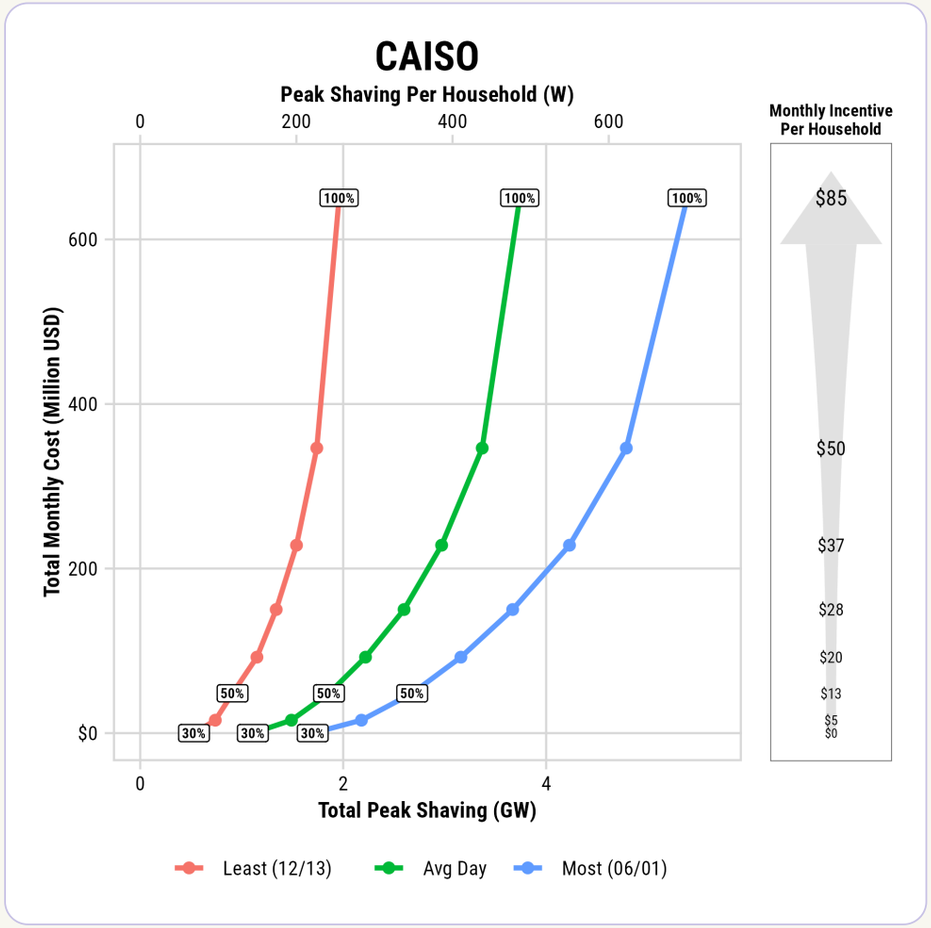
Spring	Summer	Fall	Winter
			
1.5 GW (18.4 → 16.9)	1.6 GW (24.6 → 23.0)	1.5 GW (19.8 → 18.3)	1.4 GW (22.1 → 20.7)
8.2%	6.5%	7.6%	6.3%

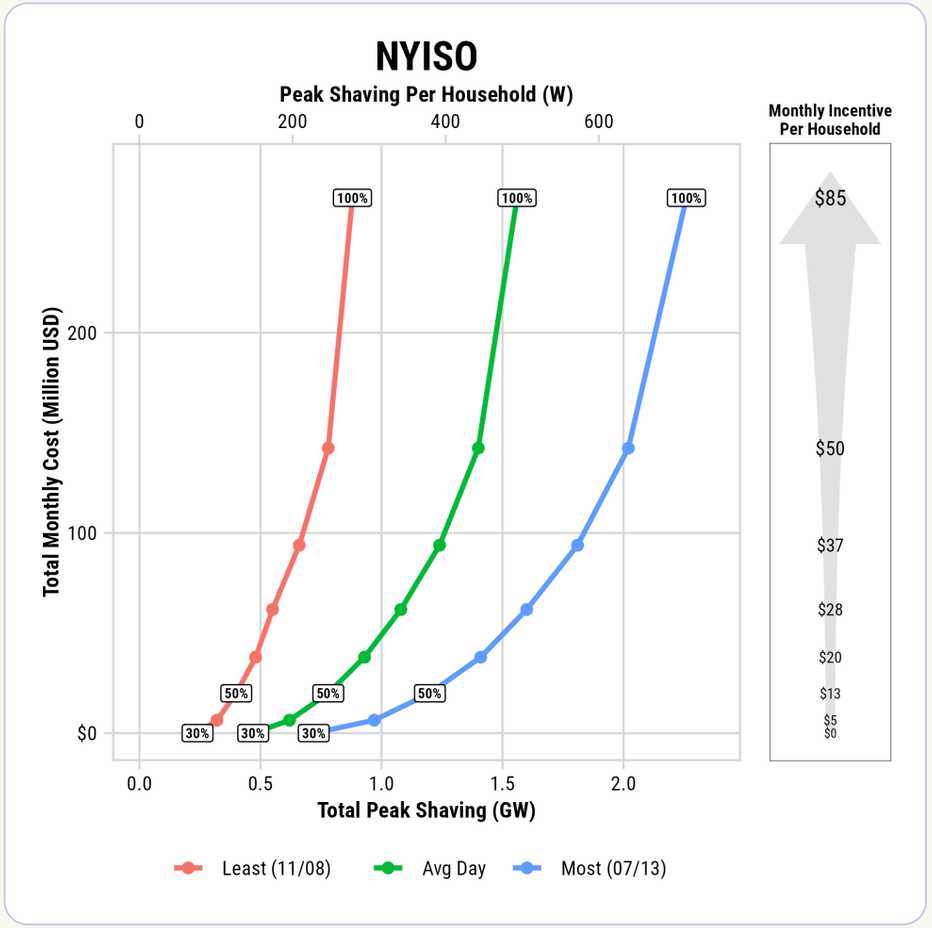
Peak shave for all seasons: **1.4–1.6 GW (6.3–8.2%)**

Peak from BEV charging decreases with higher enrollment



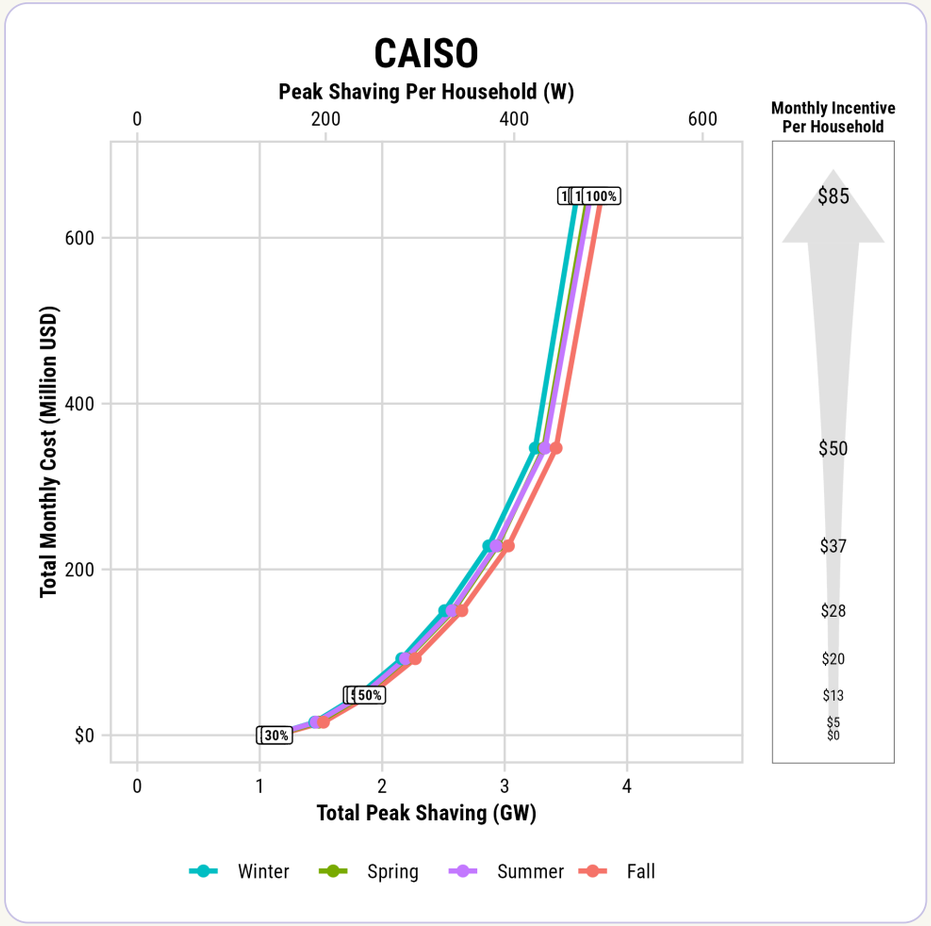
Cost Efficiency Frontier

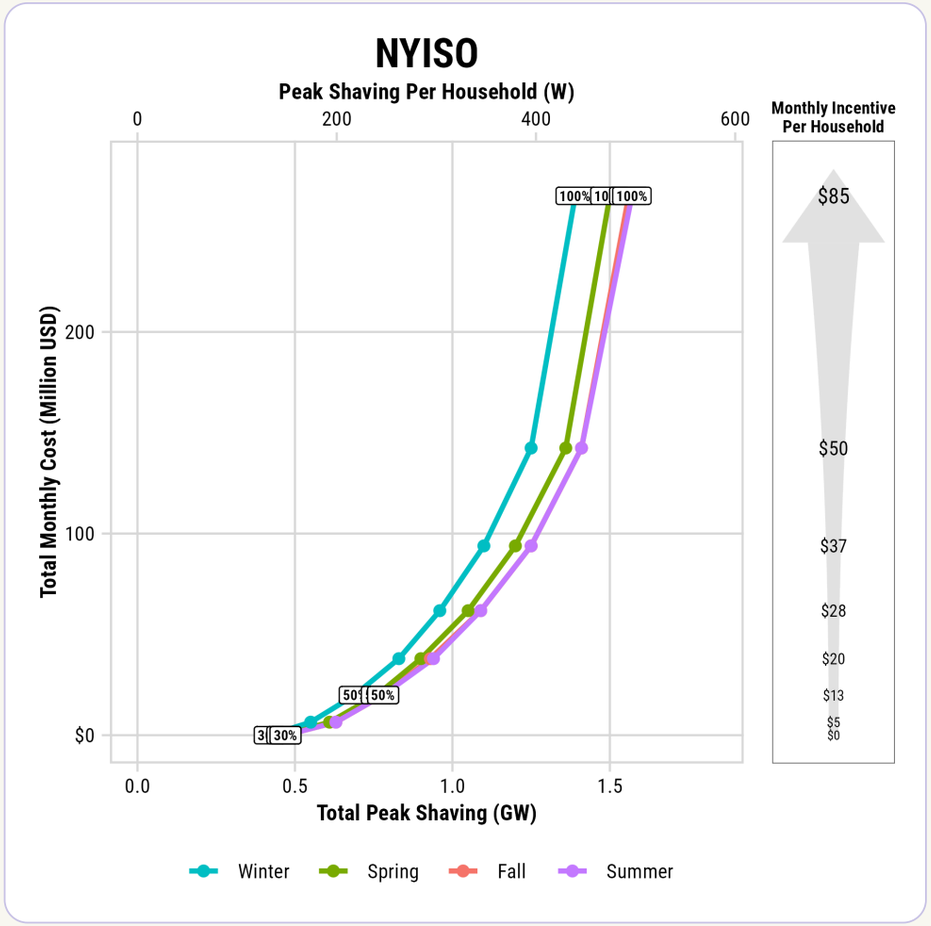




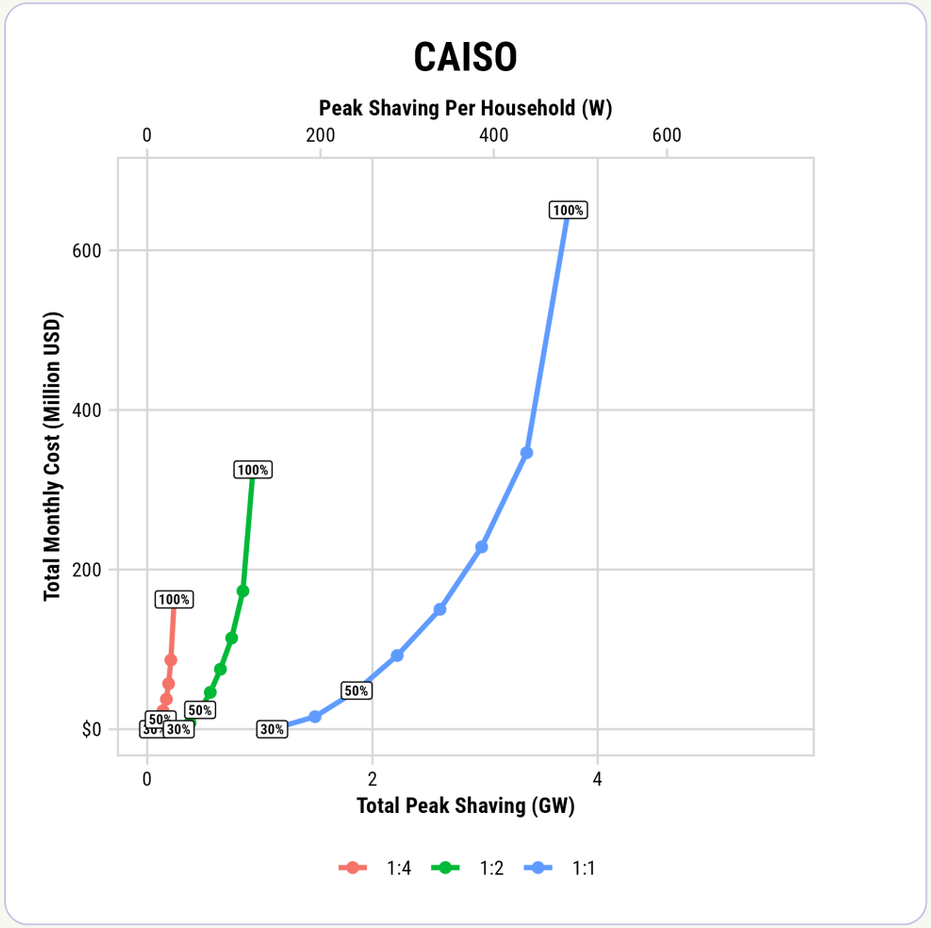
Cost efficiency is highest at moderate enrollment (30–50%).

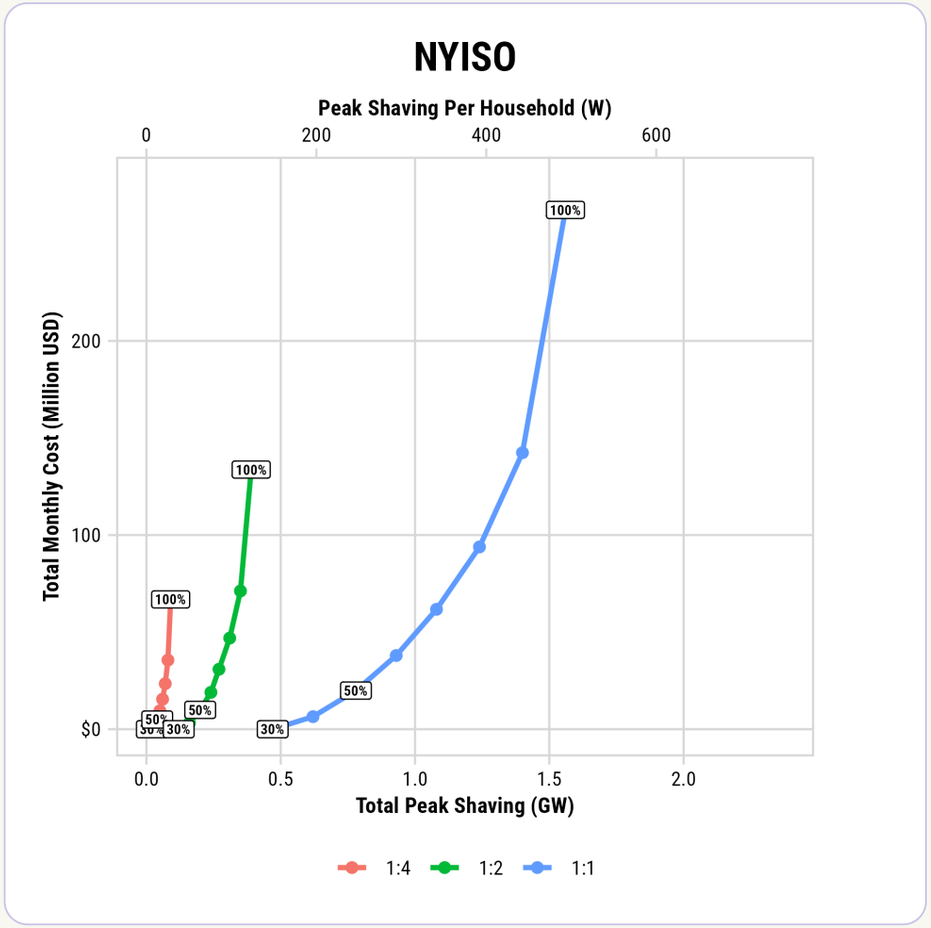
Seasonal cost efficiency is consistent in CAISO, but varies in NYISO.



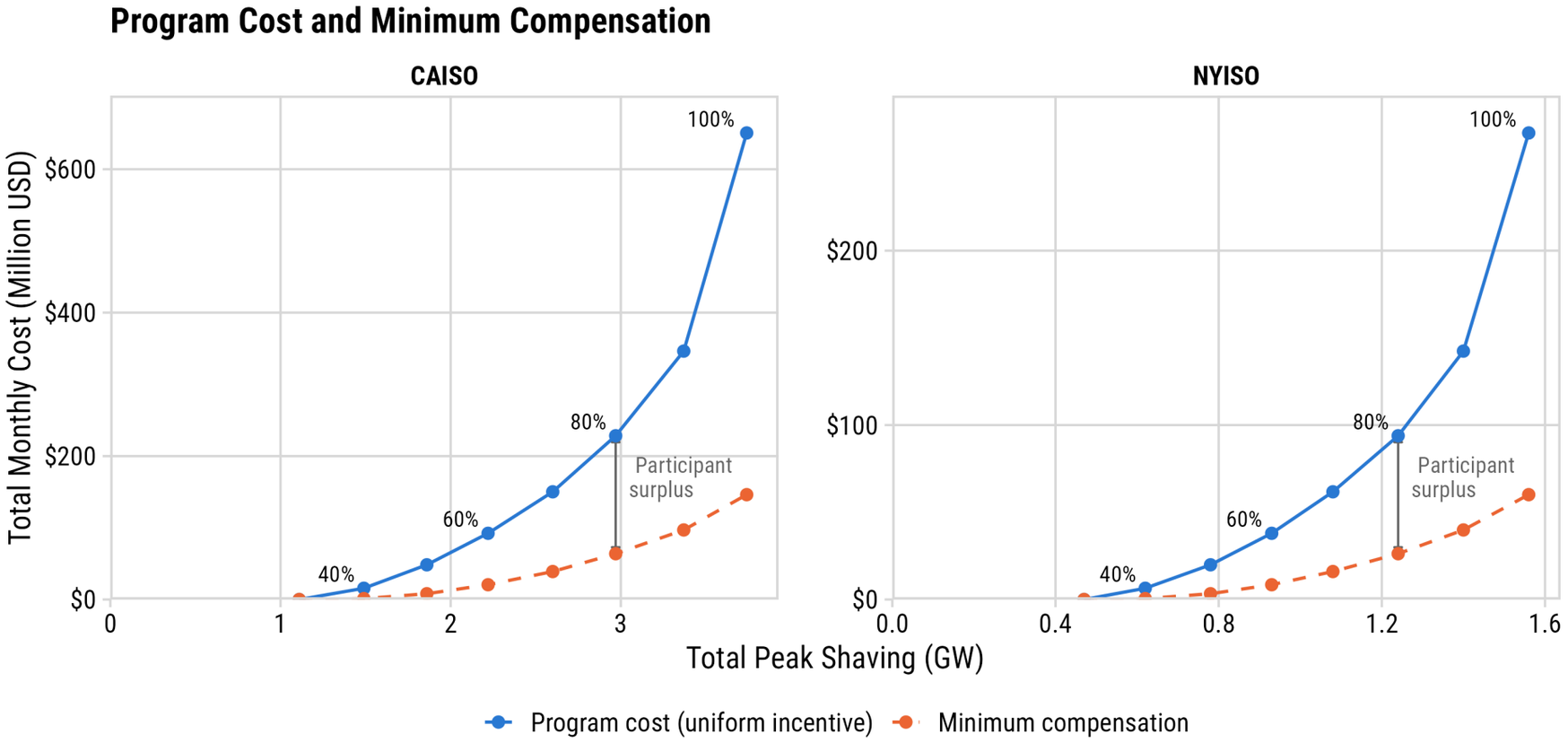


Cost efficiency across BEV adoption rates





Most program cost is participant surplus



72–92% of uniform payments exceed what enrolled households would accept.

How Do Pilot Programs Compare?

Program	Incentive	Enrollment
PCE Managed Charging (CAISO)	$0 → $40 per month	1.4% → 10.6%
MCE Sync (CAISO)	$50 bonus + $10 per month	10% of serviceable market
SmartCharge New York (NYISO)	Up to $400 per year	~7% of registered EVs (731 vehicles)

- Smart charging programs motivated EV owners to **charge off-peak.**
- More incentive results in more enrollment.
- Stated preference surveys are not calibrated against current pilot programs.



Conclusions

1. **Peak-shaving** potential is:
   up to 14.4% (5.4 GW) for CAISO, and 7.9% (2.2 GW) for NYISO.

2. The **cost-effectiveness** of peak shaving requires:
   moderate enrollment (30–50%), and seasonal effect consideration.

3. Most program cost is **participant surplus** (72–92%):
   better-targeted incentives could deliver the same shaving for less.

Limitations	a. Stated preference might be optimistic. b. Only considered single-family homes. c. Grid conditions based on 2025 data.

Acknowledgments

CO-AUTHORS

Brian Tarroja, Matthew Dean, Kate Forrest (UC Irvine) · Eric Hittinger (RIT) · Alan Jenn (UC Davis) · John Paul Helveston (GWU)

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Southern California Edison · Pacific Gas & Electric · National Grid

DATA & CODE

github.com/pingfan-hu/2025-grid-peak-shaving-public

ALFRED P. SLOAN FOUNDATION | Helveston Lab

Thanks for listening!



Pingfan Hu
George Washington University

pingfan.org

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